

Tail-sitting VTOL UAV Project

Due: 23 Jan 2026

Assignment Overview

Your tasks



Mission & Requirements

Define payload, range & autopilot



Configuration

Select single/dual/quad, vectored vs non-vectored



Structure & Aero

Design airframe & perform FEA/CFD



Propulsion

Size motors, props & ensure thrust $\geq 2 \times$ weight



Control & Simulation

Tune autopilot & run simulations



Prototype & Testing

Build, integrate & validate flight

Assessment

Criteria	Weight
Requirements & Concept	25%
Structure & Analysis	35%
Propulsion & Simulation	20%
Control & Testing	15%
Report & Presentation	05%

Resources: ArduPilot & PX4 guides, research on composites, demonstration video

Minimum to pass: complete the structural analysis stage.